Where the Movement Penalty Goes: a Correction and a Measured Boundary in Online Conformal Prediction

Author Name
Affiliation
Address
email@example.com

Abstract

Conformal PID states long-run coverage for an arbitrary bounded scorecaster, and a published corollary extends it to any predictable perturbation of the deployed threshold inside an admissible set; a contrary reading is in print, by the same authors, and is acted on. We join the halves and claim neither. What we add is a measurement outside that set: a one-scalar movement penalty on the *completed* threshold forfeits Proposition 2's rate by $42\times$ at $T = 10^6$ and, past a measurable width, returns realised miscoverage 1.000000 against $\alpha = 0.1$ while the saturation condition fires on every round. That width is $\tau^\star = \sup r_t + \sup \hat{q}_t - b/2$, exactly the corollary's admissible radius at the null scorecaster—so the conceded result is tight there, and leaving the set costs coverage rather than the rate. Under a legal non-zero scorecaster, or Conformal PID's own unbounded integrator, it does not.

1 Introduction

An online conformal method emits a threshold before each observation and moves it afterwards. Where the interval feeds a decision carrying an incumbent state that movement is charged, and forecast stability names it, scores it and prices it [Tunc et al., 2013, Godahewa et al., 2025, Van Belle et al., 2023, Pritularga and Kourentzes, 2024, Genov et al., 2026, Van Belle et al., 2026]. Holding a forecast's level fixed and comparing what its temporal structure does was invented earlier, in forecast verification, where the predictive marginal is matched on real producers and only the dependence varies [Gneiting et al., 2007, Pinson and Girard, 2012, Worsnop et al., 2018]; that coverage and mean length under-describe a deployed interval is established too [Min et al., 2026, Vaze, 2026]. None of it is claimed here.

The record contradicts itself, and the halves sit in different papers. Conformal PID [Angelopoulos et al., 2023] adds a *scorecaster* $\hat{q}_{t+1}$ to a saturating error integrator and asks of it only that it be predictable and lie in $[-b/2, b/2]$ — "any function of the past" (Theorem 1) — so a movement penalty placed in that slot is already quantified over. Also in print, of the same method: "[s]ince using a scorecaster (D-part) in C-PID breaks the theoretical coverage guarantee, we only use the quantile tracker and error integrator" [Li and Rodríguez, 2024] — one sentence in the baselines paragraph of that paper's experiments, stated and acted on there, and refereed: it survives verbatim on printed page 18443. The general statement is in print too, twice and with proof: Li et al. [2025, Corollary 2] add an arbitrary predictable d_t with $|d_t| \leq \mu_t(b/2 - |\hat{q}_t|)$ to the deployed threshold, observe that the result "has exactly the CPID error-integration form", and conclude $|E_T| \leq c\, h(T) + 1 = o(T)$; the same generic-slot argument carries AcMCP's long-run coverage [Wang and Hyndman, 2024]. The first and last authors of Li et al. [2025] are the authors of Li and Rodríguez [2024] and cite it, so the halves already share a bibliography; this is ordinary self-correction. What is useful is that four surfaces — arXiv full text and abstracts, OpenReview, and the six works Semantic Scholar records as citing it — return zero works printing both. A refereed 2026 paper separately calls the choice of scorecaster model "arbitrary" and lacking "principled guidance" [Hu et al., 2026] while asserting valid coverage itself: a complaint about model selection and variance, not about validity. **We claim neither the placement nor the derivation.**

What that leaves open is the outside of the admissible set. Corollary 2 is a *retention* result: perturbations within $\mu_t(b/2 - |\hat{q}_t|)$ keep $|E_T| \leq c\,h(T) + 1$. The modification a stability-motivated designer reaches for leaves that set — a smoother on the *completed* output. Section 3 measures what that costs and where it becomes a coverage failure.

2 Setup

The producer. At time t a forecaster fit on data before t induces a score s_t, a threshold q_t is emitted before y_t is revealed, the deployed set is $C_t = \{y : s_t(x_t, y) \leq q_t\}$ and $\mathrm{err}_t = \mathbf{1}\{s_t(x_t, y_t) > q_t\}$; the target is long-run coverage, $T^{-1}\sum_{t \leq T} \mathrm{err}_t = \alpha + o(1)$, with no model on the data. Conformal PID [Angelopoulos et al., 2023] manipulates the threshold on the score scale,

$$q_{t+1} \;=\; \hat{q}_{t+1} \;+\; r_t\Big(\textstyle\sum_{i \leq t}(\mathrm{err}_i - \alpha)\Big), \tag{1}$$

their iteration (5), in which r_t saturates: condition (4) requires $r_t(x) \geq b$ for $x \geq c\,h(t)$ and $r_t(x) \leq -b$ for $x \leq -c\,h(t)$, with h nonnegative, nondecreasing, sublinear. Writing $E_t = \sum_{i \leq t}(\mathrm{err}_i - \alpha)$, Proposition 2 gives $|E_T| \leq c\,h(T) + 1$, a coverage gap of at most $(c\,h(T) + 1)/T$; boundedness is read on realised scores.

One scalar, two readouts, two places to put it. A movement penalty on a sequentially updated scalar has two standard readouts, both prior work: a quadratic cost gives linear partial adjustment towards the target, $\tilde{q}_{t+1} = (1-\lambda)\hat{q}^{\mathrm{raw}}_{t+1} + \lambda\tilde{q}_t$ [Gârleanu and Pedersen, 2013], and a proportional cost a no-trade band, moving only when the target is more than τ away and then pulling back by exactly τ [Constantinides, 1986, Davis and Norman, 1990]. The first is what Godahewa et al. [2025] publish as post-processing and Binny and Dixit [2025, Eq. 13] apply to a deployed conformal threshold. It has two distinct positions in (1): **(B)** in the $\hat{q}_{t+1}$ slot, where both readouts keep $\{\hat{q}_t\}$ in the convex hull of $\hat{q}_1$ and the raw scorecaster's range and so inside Corollary 2's admissible set; and **(A)** downstream of the completed q_{t+1}, outside it. Placement A, what a designer reaches for because the deployed number is what has to sit still, is measured here.

The quantity, and the names it already has. We report the deployed threshold's *travel*, $\mathcal{T}_H = \sum_{t=2}^{H} |q_t - q_{t-1}|$, after actuator travel in process control. The arithmetic is not new and the name is a convenience: $\sum |\Delta q|$ over a predictive distribution is the Multi-Quantile Change [Zanotti, 2026, v3, §3.4.1]; verification calls the same movement *jumpiness* [Zsótér et al., 2009], a *convergence index* [Ehret, 2010] and forecast *(in)consistency* [Pappenberger et al., 2011], all three imported by Van Belle et al. [2026]; applied control calls it IACER [Sarhadi, 2025]; and dynamic-regret online learning calls it a path length. Table 2's caption names ten across four vocabularies: every natural name is taken, which is why a fresh one is used here, for one reading only.

Harness. $\alpha = 0.1$, $b = 2$, $c = 1$, $h(t) = \log(t + 2)$, and the clipped saturator $r_t(x) = b\,\mathrm{clip}(x/(c\,h(t)), -1, 1)$, which meets condition (4) with equality. The scorecaster is null, legal and reducing (1) to the pure error integration Proposition 2 is stated about, so the readout is the only manipulated variable. Neither free choice is neutral — condition (4) is a lower bound and this saturator meets it exactly, and Theorem 1 permits scorecasters $b/2$ either side of the null one. The adversary is specified rather than tie-broken: it plays $s_t = \mathrm{clip}(\tilde{q}_t + \varepsilon, \pm b/2)$, $\varepsilon = 10^{-9}$, against the *deployed* $\tilde{q}_t$ — the $\varepsilon \downarrow 0$ reading of "sets the score at the threshold", firing the strict indicator. Legality is Theorem 1's $[-b/2, b/2]$ on both sides, so $b/2 + b/2 = b$ is exactly tight.

3 The forfeit, measured

The rate is forfeited in the time constant that stability motivates raising. A downstream partial adjustment of weight w delays the correction the integrator has already computed. Its excursion at $T = 10^4$ is $6.30/63.00/623.70$ for $w = 0.9/0.99/0.999$, and for the last two it is *flat in T* while

Table 1: Placement A: a one-scalar readout on the *completed* threshold; one deterministic adversary, one pass to $T = 10^6$, the i.i.d. control at one seed. Proposition 2's bound is 13.2061 at $T = 2{\times}10^5$ and 14.8155 at $T = 10^6$; "ratio" is $\max_t |E_t|$ over it, "Sat." the rounds on which (4) delivers its full $\pm b$.

Readout on the completed threshold	miscov. $2{\times}10^5$	$\max_t \lvert E_t \rvert$ 10^6	ratio	Sat. 10^6	travel 10^6
none (control)	0.100035	7.80	0.53	0.0000	28,311
partial adj. $w = 0.9$	0.100030	7.50	0.51	0.0000	4,081
partial adj. $w = 0.99$	0.100040	63.00	4.25	0.0007	1,024
partial adj. $w = 0.999$	0.100035	623.70	42.10	0.0097	202
growing time constant $p = 0.9$	0.100030	53.00	3.58	0.3181	16
running mean	0.099940	49.50	3.34	0.3647	9
dead band $\tau = 0.5$	0.100035	10.60	0.72	0.0000	2,045
dead band $\tau = 0.9$	0.100060	13.20	0.89	0.0038	1,051
dead band $\tau = 1.5$	**1.000000**	900,000	60,747	1.0000	0.5

the bound grows like $\log T$, so the ratio falls with the horizon, 61.08 to 42.10; the measured law across the arms is $\max_t |E_t| \approx \max\{0.5\,h(T) + 0.9,\ 0.63/(1-w)\}$. Five of the nine exceed the bound at $T = 10^6$, so the forfeit is a property of the smoothing *strength* rather than of Placement A, and it buys something: the $w = 0.999$ arm cuts deployed travel from 28,311 to 202. The sign is the opposite of the intuitive one: a readout on the completed q_{t+1} touches neither r_t nor the integrator's argument, so it cannot put condition (4) at risk; it delays the correction the integrator is waiting for, so the accumulator excurses *further*. Smoothing the output does not damp E_t, it feeds it. The 623.70 turns on the tie rule: an adversary playing the threshold *exactly* gives it with ties scored as misses and 70.80 under the strict indicator.

Coverage survives most of the time, and then it does not. Realised miscoverage at $T = 2{\times}10^5$ lies in 0.099940–0.100060 over the control and seven of the eight smoothed arms; the eighth loses coverage completely, and the width at which that happens is measurable. A dead band pulls the deployed threshold back by τ, so under saturation it is pinned τ below the integrator's ceiling; measured, its realised supremum is $b - \tau$ to the printed digit at every $\tau \geq 0.9$. The adversary's best legal play is $b/2$, so coverage is lost exactly when that supremum drops below it, at $\tau^\star = \sup_x r_t(x) + \sup_t \hat{q}_t - b/2$; bands with $\tau \leq \tau^\star$ cover. **The harness puts** $\sup r_t = b$ **and** $\hat{q} \equiv 0$**, so** $\tau^\star = b/2$ **here** — one point of the admissible class, not a property of Placement A. Under the equally legal constant $\hat{q} \equiv +b/2$ the same $\tau = 1.5$ band covers at $T = 10^6$, miscoverage 0.100010 and $\max_t |E_t| = 11.20$, inside the bound; under $\hat{q} \equiv -b/2$, also legal, $\tau^\star = 0$ and even $\tau = 0.5$ returns 1.000000. At $\hat{q} \equiv 0$ the transition is a step: $\tau = 1.000$ covers at $T = 10^5$ while $\tau = 1.001$ returns 1.000000. Past $\tau^\star$ the threshold sits permanently below the reachable score range, the loop never closes and $|E_t| = (1-\alpha)t$ grows linearly, 60,747 times the bound at $T = 10^6$; $\tau = 2.0$, 2.5, 3.0 and 5.0 fail too, so the failing set is $(\tau^\star, \infty)$, open at the left — the endpoint covers. **And the failing arms have** $\mathrm{Sat.} = 1.0000$: the integrator delivers its full $\pm b$ on every round and the failure happens anyway. Condition (4) holds by construction; what fails is Proposition 2's other hypothesis, that the threshold closing the loop *is* the integrator's output.

What that measures is that the conceded result is tight. A dead band displaces the deployed threshold by exactly τ when it releases, so "inside Li et al.'s Corollary 2" reads $\tau \leq \mu_t(b/2 - |\hat{q}_t|)$, which at $\mu_t = 1$, $\hat{q}_t \equiv 0$ is the $b/2$ that $\tau^\star$ returns. What is measured is therefore not a fresh boundary but that the published sufficient condition is *necessary* there, and that the price of leaving the set is coverage rather than the rate. Elsewhere it is only sufficient: at $\hat{q}_t \equiv +b/2$ Corollary 2 permits radius 0 while coverage survives to $\tau = b$.

4 Where this sits

Of the 862 references returned for Table 2's twenty-six anchors, 13 fall in a neighbouring literature and twelve of those sit in three forecast-stability papers: Van Belle et al. [2026], co-authored by Pierre Pinson, cite eight works of verification, Genov et al. [2026] three of smoothed online optimisation, and Zanotti [2026] build their intervals out of conformal prediction. The edge exists where an author bridges it, and the cross-citation is concentrated rather than sparse. **We claim no disconnection**, and the counts above do not support one: the arXiv full-text index that returns zero for the switching-cost method names paired with "conformal" over-matches phrases, so its zeros are conservative and its

Table 2: Four literatures name the movement of a sequentially updated forecast in four vocabularies: *abrupt threshold changes*; *forecast (in)stability* (MASC, MAC/SDC, MQC/SMQC, congruence, nervousness); *jumpiness*, *convergence index*, *(in)consistency*; and *switching cost*. Cross-citation from Semantic Scholar reference lists, 20 August 2026, over that column's anchors.

	Online conformal	**Forecast stability**	**Verification**	**Switching-cost OL**
Has that the others lack	a distribution-free long-run-coverage guarantee for *any* bounded scorecaster, with a rate [Angelopoulos et al., 2023, Thm 1, Prop. 2]—and an admissible set measured here to be tight	the scored movement functional, the readout map as post-processing [Godahewa et al., 2025, Eq. 3], and a priced decision [Genov et al., 2026, §4.2], [Van Belle et al., 2026]	the matched-level design on real producers from 2012, with a control *stronger* than the pair: the whole marginal [Pinson and Girard, 2012, Worsnop et al., 2018]	worst-case theory for the penalised problem: Thm 2, and a Thm 7 trade-off in which sublinear regret is bought by growing θ with T, which grows the ratio [Andrew et al., 2013]
Lacks that the others have	a calendar-time path functional and a decision charged to *move* rather than to violate; its metric is "zero for deterministic methods by design" [Min et al., 2026, §4]	a coverage object in its canonical works: *conformal*, *coverage*, *prediction interval* occur 0 times in Godahewa et al. [2025], Genov et al. [2026], Van Belle et al. [2026]; Zanotti [2026] excepted	a decision that pays to move—commissioned and declined [Pinson and Girard, 2012]—and the quantity itself: matched marginals fix the width path	any interval or coverage object; regret buys marginal coverage i.i.d. but not adversarially [Ramalingam et al., 2025]
Cites the other three?	$0/437$, $1/437$, $0/437$; cites the shared OCO ancestor [Zinkevich, 2003]	$1/226$, $8/226$, $3/226$—all in three papers	$0/106$ in every direction available	$0/93$ in every direction available

positives are not counts, and an exact instrument refutes the reading they invite — of the 875 works citing Kalai and Vempala [2005], two are online conformal, Chen et al. [2023], cited here, and Zhang et al. [2024]. Four fields name one quantity many ways; the object that settles its validity sits in one.

What is conceded, by name. The placement is prior work: Angelopoulos et al. [2023] state coverage for any admissible integrator and any scorecaster and run a smoothing-family model in that slot themselves; Dupuy et al. [2026] give the generic argument in their Appendix A and Duerst et al. [2024] adopt that scorecaster. The derivation is prior work too, and that is the concession that matters most: Li et al.'s Corollary 2 proves the retention statement in Conformal PID's own notation [Li et al., 2025], with Wang and Hyndman [2024] an independent second instance. The smoother object, the metric arithmetic and both readout maps are conceded in Section 2, and Zheng et al. [2025], Wu et al. [2025], Chen et al. [2023], Lade et al. [2026] each place a smoother inside their own validity argument. One vocabulary out, this is integrator anti-windup with a feedforward path, where free-until-the-bound-binds is the defining specification rather than a finding [Galeani et al., 2009]. Nearest in the other direction, Gao et al. [2025] run the same recursion but gate the *model* update on the conformal score; their one non-standard assumption is that $\sum_{t\le T} q_t \le O(\alpha\, h(T))$ for a sublinear h, supported with a plot and left to later work to derive. That accumulation is what Section 3 measures diverging.

5 Limitations

One construction, one adversary, one saturator — and its *level* is the operative choice. $0.63/(1-w)$ is specific to that level, not to the saturator's shape: raising the level to $4b$ — admissible, since (4) is a lower bound — cuts 623.70 to 120.60, and under the unbounded tangent integrator Angelopoulos et al. [2023] use in all their own experiments no width tried loses coverage, $\tau = 1.5$ covering at 0.100001 and 623.70 falling to 20.10. Both results are about *bounded* admissible integrators, which is what (4) permits and not what its authors run. **The inherited guarantee is weak where it applies.** With that integrator, $h(t) = t/\log t$ and their Appendix C constants (K_I at the score bound), the bound decays as $O(1/\log T)$: at $T = 2500$, our horizon, it certifies coverage only into $[0.821, 0.979]$ against 0.90. What is forfeited was not a tight certificate. **The dead-band widths are absolute, and the failure is total only against the adversary.** τ is in score units with $b = 2$, so the failing $\tau = 1.5$ is $0.75\,b$; under i.i.d. scores that arm returns 0.249376, under the tangent 0.100025. What survives is the mechanism: a band wider than $\tau^\star$ caps the reachable threshold below the reachable score range. **The harness was rebuilt against numbers already read**, which carries less evidential weight than a blind rerun; the free choices are registered in the code and travel with every output. **Seven of the twenty-six anchors return an empty reference list**, so Table 2's counts are gaps rather than zeros. **No downstream readout here recovers the bound.**

References

Lachlan L. H. Andrew, Siddharth Barman, Katrina Ligett, Minghong Lin, Adam Meyerson, Alan Roytman, and Adam Wierman. A tale of two metrics: Simultaneous bounds on competitiveness and regret. In *Proceedings of the 26th Conference on Learning Theory (COLT 2013)*, 2013.

Anastasios N. Angelopoulos, Emmanuel J. Candès, and Ryan J. Tibshirani. Conformal pid control for time series prediction. In *Advances in Neural Information Processing Systems 36 (NeurIPS 2023)*, 2023. doi: 10.52202/075280-1000.

Allen Emmanuel Binny and Anushri Dixit. Who moved my distribution? conformal prediction for interactive multi-agent systems, 2025.

Jiadong Chen, Yang Luo, Xiuqi Huang, Fuxin Jiang, Yangguang Shi, Tieying Zhang, and Xiaofeng Gao. IPOC: An adaptive interval prediction model based on online chasing and conformal inference for large-scale systems. In *Proceedings of the 29th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*, pages 202–212, 2023. doi: 10.1145/3580305.3599396.

George M. Constantinides. Capital market equilibrium with transaction costs. *Journal of Political Economy*, 94(4):842–862, 1986. doi: 10.1086/261410.

M. H. A. Davis and A. R. Norman. Portfolio selection with transaction costs. *Mathematics of Operations Research*, 15(4):676–713, 1990. doi: 10.1287/moor.15.4.676.

Ricarda Duerst, Jonas Schöley, Julia Hellstrand, and Mikko Myrskylä. Calibrating probabilistic forecast paths on past forecast errors: an application to the Finnish total fertility rate. Technical Report WP-2024-016, Max Planck Institute for Demographic Research, 7 2024.

Théo Dupuy, Binbin Xu, Stéphane Perrey, Jacky Montmain, and Abdelhak Imoussaten. Relevance-aware thresholding in online conformal prediction for time series. In *Communications in Computer and Information Science*, pages 196–217, 2026. doi: 10.1007/978-3-032-16708-8_17.

Uwe Ehret. Convergence index: a new performance measure for the temporal stability of operational rainfall forecasts. *Meteorologische Zeitschrift*, 19(5):441–451, 2010. doi: 10.1127/0941-2948/2010/0480.

Sergio Galeani, Sophie Tarbouriech, Matthew Turner, and Luca Zaccarian. A tutorial on modern anti-windup design. *European Journal of Control*, 15(3-4):418–440, 2009. doi: 10.3166/ejc.15.418-440.

Ben Gao, Jordan Patracone, Stéphane Chrétien, and Olivier Alata. Conformal online learning of deep Koopman linear embeddings, 2025.

Nicolae Gârleanu and Lasse Heje Pedersen. Dynamic trading with predictable returns and transaction costs. *The Journal of Finance*, 68(6):2309–2340, 2013. doi: 10.1111/jofi.12080.

Evgenii Genov, Julian Ruddick, Christoph Bergmeir, Majid Vafaeipour, Thierry Coosemans, Salvador García, and Maarten Messagie. Balancing forecast accuracy and switching costs in online optimization of energy management systems. *Expert Systems with Applications*, 298:129305, 2026. doi: 10.1016/j.eswa.2025.129305.

Tilmann Gneiting, Fadoua Balabdaoui, and Adrian E. Raftery. Probabilistic forecasts, calibration and sharpness. *Journal of the Royal Statistical Society Series B: Statistical Methodology*, 69(2):243–268, 2007. doi: 10.1111/j.1467-9868.2007.00587.x.

Rakshitha Godahewa, Christoph Bergmeir, Zeynep Erkin Baz, Chengjun Zhu, Zhangdi Song, Salvador García, and Dario Benavides. On forecast stability. *International Journal of Forecasting*, 41(4):1539–1558, 2025. doi: 10.1016/j.ijforecast.2025.01.006.

Dongjian Hu, Junxi Wu, Shu-Tao Xia, and Changliang Zou. Distribution-informed online conformal prediction, 2026.

Adam Kalai and Santosh Vempala. Efficient algorithms for online decision problems. *Journal of Computer and System Sciences*, 71(3):291–307, 2005. doi: 10.1016/j.jcss.2004.10.016.

Ankit Lade, Sai Krishna J., and Indar Kumar. Bias-corrected adaptive conformal inference for multi-horizon time series forecasting, 2026.

Ruipu Li and Alexander Rodríguez. Neural conformal control for time series forecasting, 2024.

Ruipu Li, Daniel Menacho, and Alexander Rodríguez. Optimization-based online conformal prediction for multi-step forecasting, 2025.

Yizhou Min, Yizhou Lu, Lanqi Li, Zhen Zhang, and Jiaye Teng. Questioning the coverage-length metric in conformal prediction: When shorter intervals are not better, 2026.

F. Pappenberger, H. L. Cloke, A. Persson, and D. Demeritt. HESS Opinions "on forecast (in)consistency in a hydro-meteorological chain: curse or blessing?". *Hydrology and Earth System Sciences*, 15(7):2391–2400, 2011. doi: 10.5194/hess-15-2391-2011.

P. Pinson and R. Girard. Evaluating the quality of scenarios of short-term wind power generation. *Applied Energy*, 96:12–20, 2012. doi: 10.1016/j.apenergy.2011.11.004.

Kandrika Pritularga and Nikolaos Kourentzes. Forecast congruence: A quantity to align forecasts and inventory decisions. SSRN working paper 4711817, 2024.

Ramya Ramalingam, Shayan Kiyani, and Aaron Roth. The relationship between no-regret learning and online conformal prediction. In *Proceedings of the 42nd International Conference on Machine Learning (ICML 2025)*, 2025.

Pouria Sarhadi. On the standard performance criteria for applied control design: PID, MPC or machine learning controller?, 2025.

Huseyin Tunc, Onur A. Kilic, S. Armagan Tarim, and Burak Eksioglu. A simple approach for assessing the cost of system nervousness. *International Journal of Production Economics*, 141(2): 619–625, 2013. doi: 10.1016/j.ijpe.2012.09.022.

Jente Van Belle, Ruben Crevits, and Wouter Verbeke. Improving forecast stability using deep learning. *International Journal of Forecasting*, 39(3):1333–1350, 2023. doi: 10.1016/j.ijforecast.2022.06.007.

Jente Van Belle, Honglin Wen, Wouter Verbeke, and Pierre Pinson. Stabilizing distribution-free probabilistic forecasts, 2026.

Rahul Vaze. Simultaneous coverage and efficiency guarantee in online conformal prediction, 2026.

Xiaoqian Wang and Rob J. Hyndman. Online conformal inference for multi-step time series forecasting, 2024.

Rochelle P. Worsnop, Michael Scheuerer, Thomas M. Hamill, and Julie K. Lundquist. Generating wind power scenarios for probabilistic ramp event prediction using multivariate statistical post-processing. *Wind Energy Science*, 3(1):371–393, 2018. doi: 10.5194/wes-3-371-2018.

Junxi Wu, Dongjian Hu, Yajie Bao, Shu-Tao Xia, and Changliang Zou. Error-quantified conformal inference for time series, 2025.

Marco Zanotti. Analyzing the retraining frequency of global forecasting models: towards more stable forecasting systems, 2026.

Zhiyu Zhang, Zhou Lu, and Heng Yang. The benefit of being Bayesian in online conformal prediction, 2024.

Mingyi Zheng, Hongyu Jiang, Yizhou Lu, and Jiaye Teng. Smoothing-based conformal prediction for balancing efficiency and interpretability, 2025.

Martin Zinkevich. Online convex programming and generalized infinitesimal gradient ascent. In *Proceedings of the 20th International Conference on Machine Learning (ICML 2003)*, 2003.

Ervin Zsótér, Roberto Buizza, and David Richardson. "Jumpiness" of the ECMWF and Met Office EPS control and ensemble-mean forecasts. *Monthly Weather Review*, 137(11):3823–3836, 2009. doi: 10.1175/2009MWR2960.1.